# WIRESHARK

## Network Protocol Analysis

*Laboratory Exercises for Beginners*

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**Lab Section: Wireshark 1**

**Protocols Covered:**

| HTTP | ICMP | DNS | TCP | ARP |
|---|---|---|---|---|

# Introduction to Wireshark

### What is Wireshark?

Wireshark is a free, open-source network protocol analyzer. It captures network packets in real time and lets you inspect them in detail — like a microscope for your network traffic.

In these exercises, you will use Wireshark to observe real network communication, understand how different protocols work, and analyze packet-level data.

## Learning Objectives

- Install and navigate the Wireshark interface
- Capture live network traffic using filters
- Analyse HTTP, ICMP, DNS, TCP, and ARP packets
- Read and interpret packet headers and payloads
- Answer analysis questions and attach supporting screenshots

## Software Requirements

- Wireshark 4.x (download from https://www.wireshark.org/download.html)
- Windows 10/11, macOS, or Linux
- Active internet connection
- Administrator / sudo privileges on your machine

### □ Submission Instructions

For every question marked with □, capture a screenshot from Wireshark that shows the relevant traffic.
Paste each screenshot directly into this document in the space provided.
Write short written answers where the question asks for one.
Submit the completed document as a .docx or .pdf file via your course portal.

# Part 0 — Getting Started with Wireshark

## 0.1 Installing Wireshark

1. Go to https://www.wireshark.org/download.html
2. Download the installer for your operating system.
3. Run the installer and accept all default settings (including Npcap on Windows).
4. Launch Wireshark after installation.

## 0.2 Wireshark Interface Overview

| UI Area | Description |
|---|---|
| **Packet List Pane** | Top panel — shows all captured packets in chronological order (No., Time, Source, Destination, Protocol, Info). |
| **Packet Details Pane** | Middle panel — shows the decoded structure of the selected packet, organized by protocol layer. |
| **Packet Bytes Pane** | Bottom panel — shows the raw hexadecimal and ASCII bytes of the selected packet. |
| **Display Filter Bar** | Bar at the top — type filters here to show only specific packets (e.g., http, icmp, dns). |
| **Capture Interfaces** | Start screen — lists available network interfaces (Wi-Fi, Ethernet, Loopback, etc.). |

## 0.3 Starting a Capture

**1** **Open Wireshark**
You will see a list of network interfaces on the start screen.

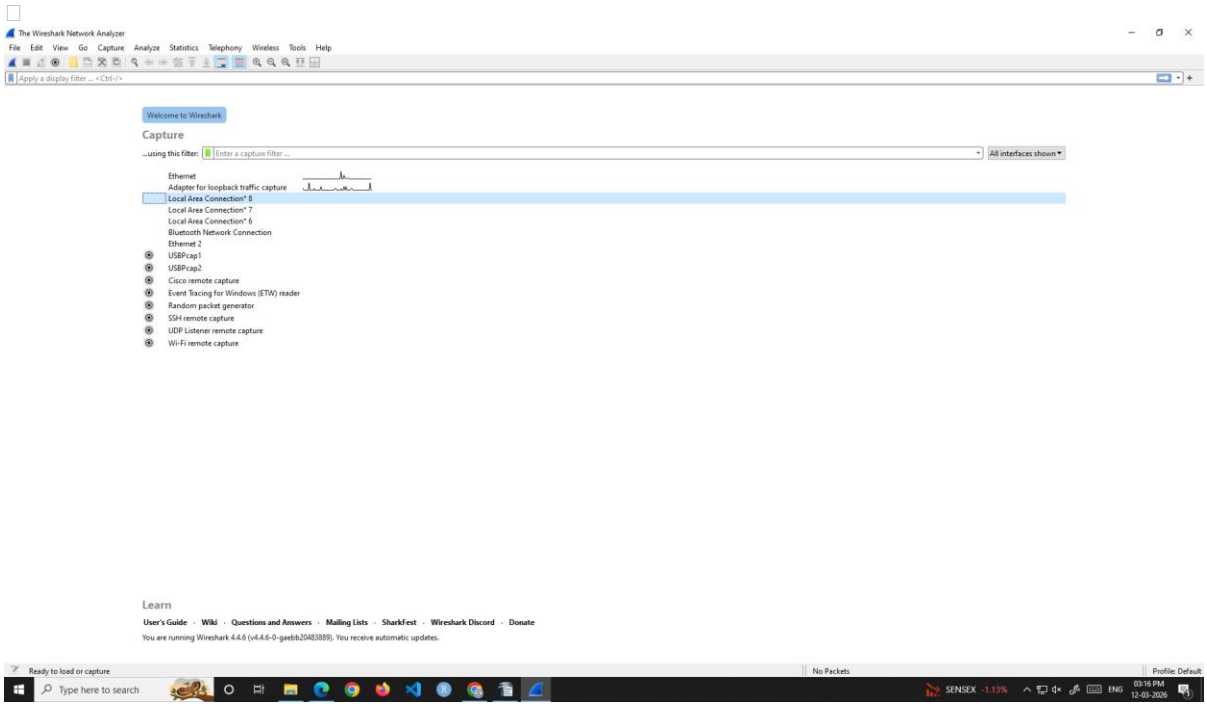
**2** **Choose your active interface**
Select the interface that is currently active — usually Wi-Fi or Ethernet. You will see a live graph of traffic activity next to active interfaces.

**3** **Double-click the interface**
This immediately starts capturing all network packets on that interface.

**4** **Stop the capture**
Click the red Stop button (■) in the toolbar, or press Ctrl+E (Cmd+E on Mac).

**5** **Use display filters**
In the filter bar, type a protocol name like http or icmp and press Enter to show only those packets.

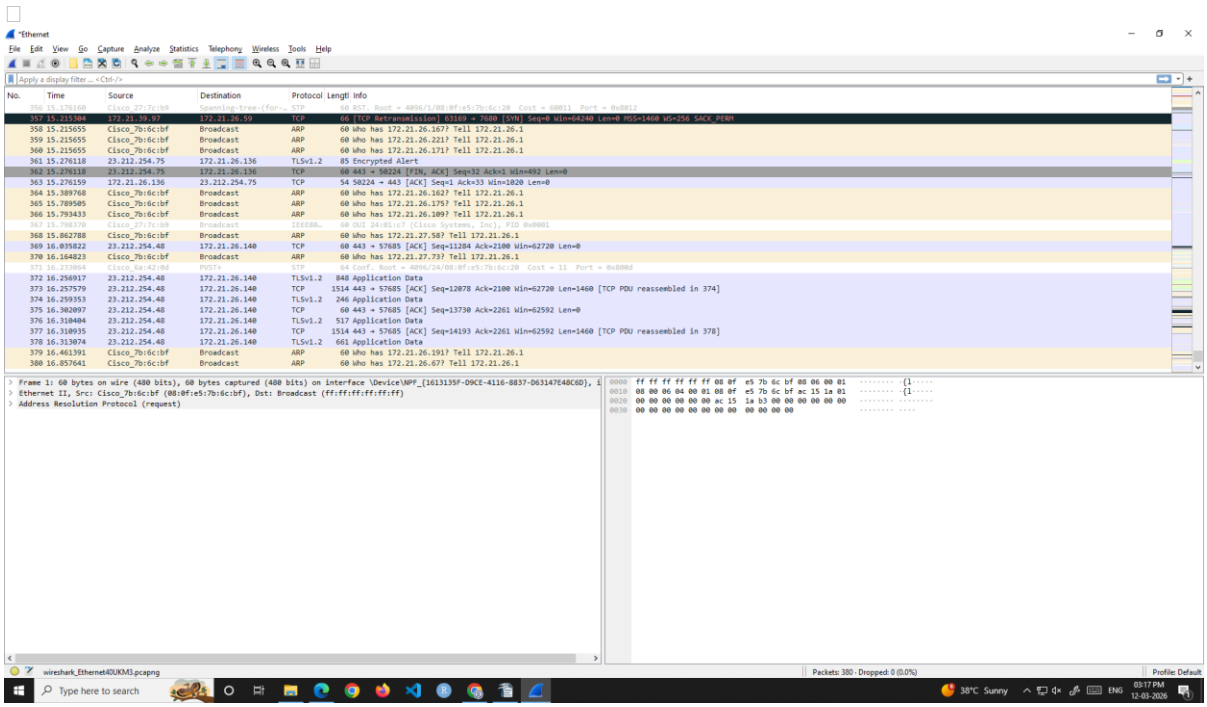
**Question 1:** Take a screenshot of the Wireshark home screen showing the list of available network interfaces on your computer.




*Attach Screenshot Here*

**Your Answer / Observation:**

**Question 2:** Start a capture on your active interface, let it run for 10 seconds, then stop it. Take a screenshot of the captured packet list. How many packets were captured? Write the number in the answer box.

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**Your Answer / Observation:**

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# Exercise 1 — ICMP (Internet Control Message Protocol)

### What is ICMP?

ICMP (Internet Control Message Protocol) is used by network devices to send error messages and operational information.
The most familiar use of ICMP is the ping command — it sends ICMP Echo Request packets and waits for Echo Reply packets.
ICMP operates at the Network Layer (Layer 3) of the OSI model.

Key ICMP message types:

- Type 0 — Echo Reply
- Type 3 — Destination Unreachable
- Type 8 — Echo Request
- Type 11 — Time Exceeded

## Steps — Capturing ICMP Traffic

**1** **Start a new capture**
Open Wireshark and start capturing on your active network interface.

**2** **Open a terminal / command prompt**
On Windows: press Win+R, type cmd, press Enter. On Mac/Linux: open Terminal.

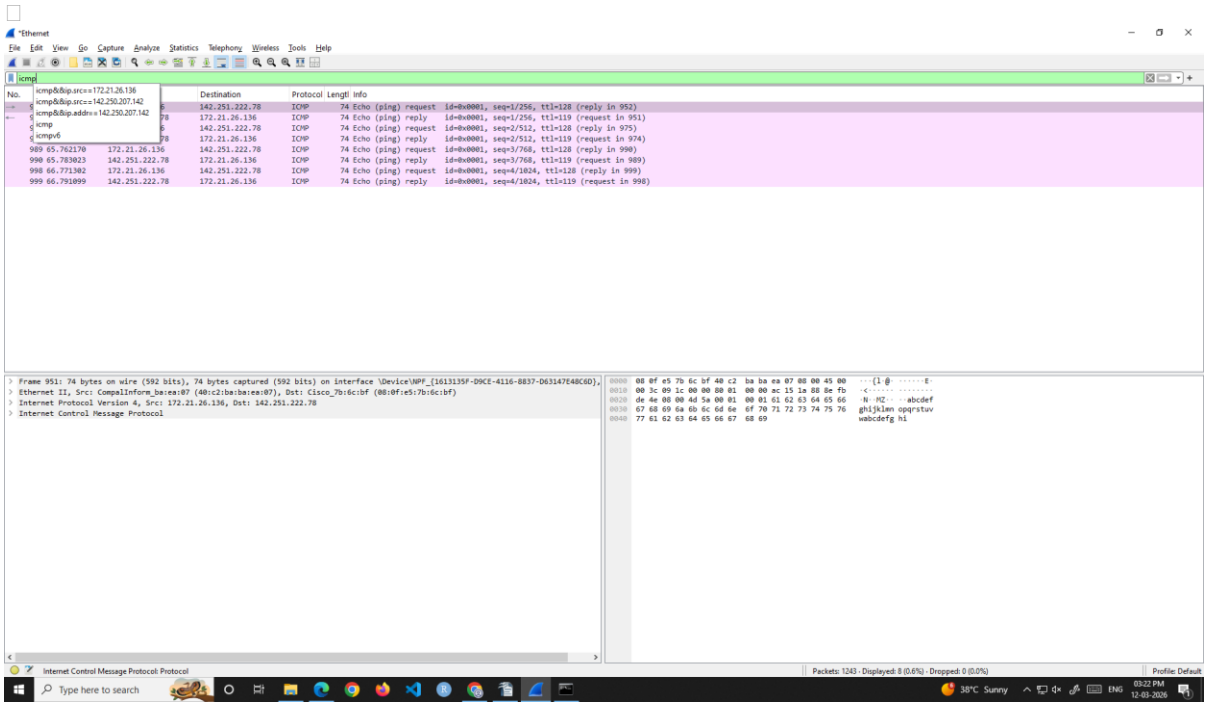
**3** **Run the ping command**
Type the following and press Enter: Windows: ping google.com Mac/Linux: ping -c 4 google.com

**4** **Return to Wireshark and stop the capture**
Click the Stop button (■).

**5** **Apply a display filter**
Type icmp in the filter bar and press Enter. You should now see only ICMP packets.

## Questions — ICMP Analysis

**Question 3:** Apply the icmp filter and take a screenshot showing the ICMP packets in the Packet List pane. You should see alternating Echo Request and Echo Reply packets.




*Attach Screenshot Here*

**Your Answer / Observation:**

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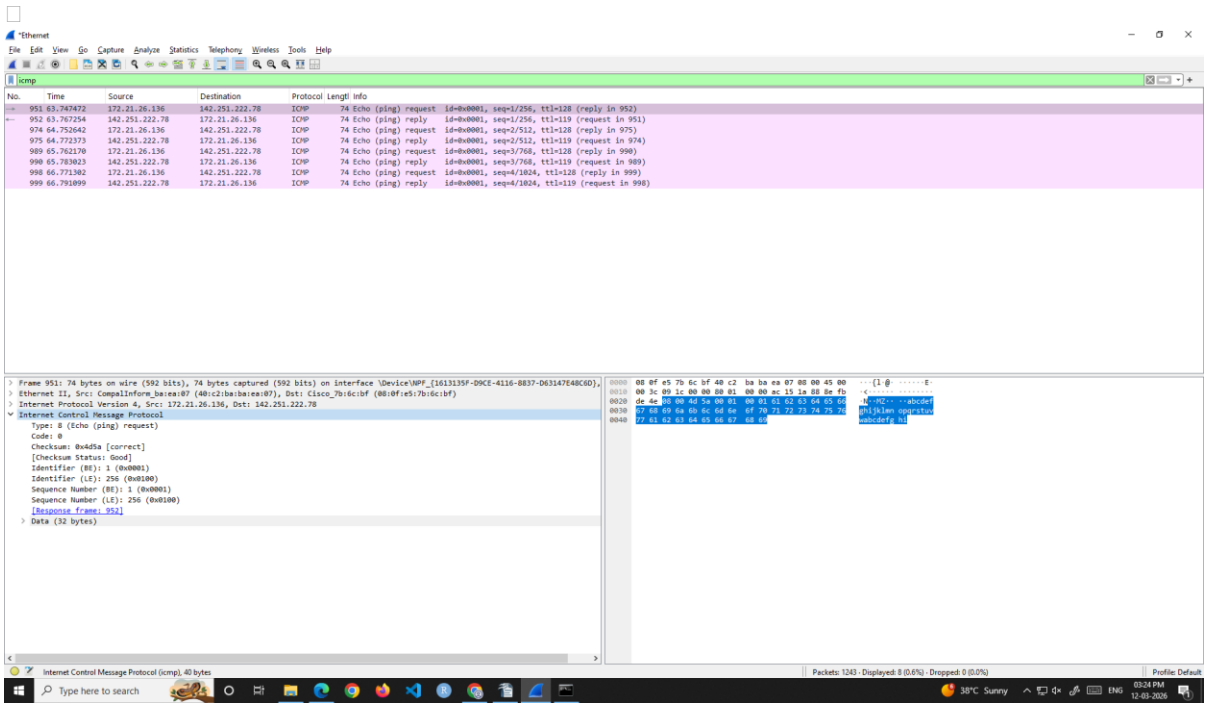
**Question 4:** Click on one Echo Request packet. In the Packet Details pane, expand the 'Internet

Control Message Protocol' layer. Take a screenshot. What is the Type value and what does it mean?

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**Your Answer / Observation:**

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**Question 5:** Click on one Echo Reply packet and expand the ICMP layer. Take a screenshot. What is the Type value for a Reply?

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 6:** Look at the Source and Destination columns for the ICMP packets. What is the IP address of google.com that responded to your ping? Write it below.

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 7:** In the Packet Details pane, expand the 'Internet Protocol Version 4' layer. Take a screenshot showing the TTL (Time to Live) field. What is the TTL value of the Echo Reply you received?

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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### What is TTL?

TTL (Time to Live) is a field in the IP header that limits how many router hops a packet can make. Each router decrements TTL by 1. When TTL reaches 0, the packet is discarded and an ICMP Time Exceeded message is sent back.
Common starting TTL values: Windows = 128, Linux/Mac = 64, Cisco routers = 255.

# Exercise 2 — HTTP (Hypertext Transfer Protocol)

### What is HTTP?

HTTP (Hypertext Transfer Protocol) is the foundation of data communication on the World Wide Web. When you open a webpage, your browser sends an HTTP GET request to the server, and the server responds with an HTTP 200 OK response containing the page content.
HTTP operates at the Application Layer (Layer 7) of the OSI model and typically uses TCP port 80.

Common HTTP methods: GET, POST, PUT, DELETE
Common status codes: 200 OK | 301 Redirect | 404 Not Found | 500 Server Error

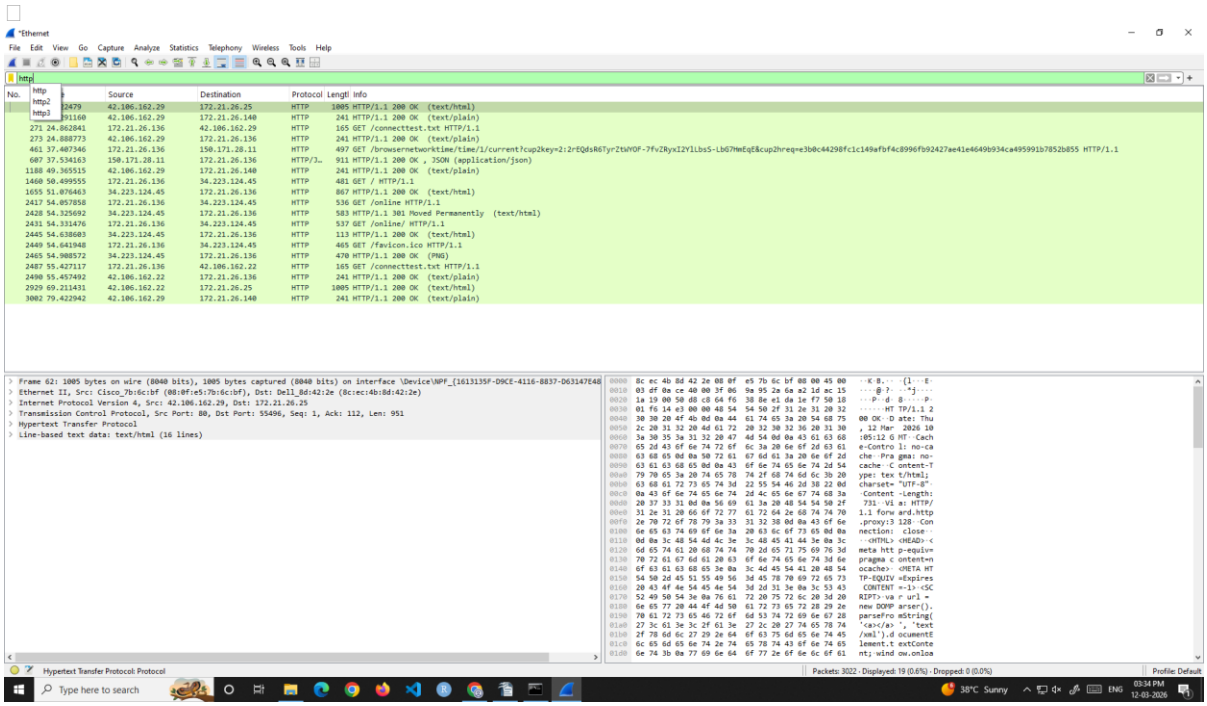
Note: Most modern websites use HTTPS (encrypted). We will use a plain HTTP test site for this exercise.

## Steps — Capturing HTTP Traffic

| | |
|---|---|
| 1 | **Start a new capture in Wireshark**<br>Start capturing on your active network interface. |
| 2 | **Open a web browser**<br>Open Chrome, Firefox, or Edge. |
| 3 | **Visit an HTTP test site**<br>Navigate to: http://neverssl.com(This site does NOT use HTTPS, so Wireshark can see the full request.) |
| 4 | **Wait for the page to load, then return to Wireshark**<br>Stop the capture. |
| 5 | **Apply the http filter**<br>Type http in the filter bar and press Enter. |

## Questions — HTTP Analysis

**Question 8:** After applying the http filter, take a screenshot of the Packet List pane. You should see GET requests and HTTP responses. Identify the first GET request in the list.

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**Your Answer / Observation:**

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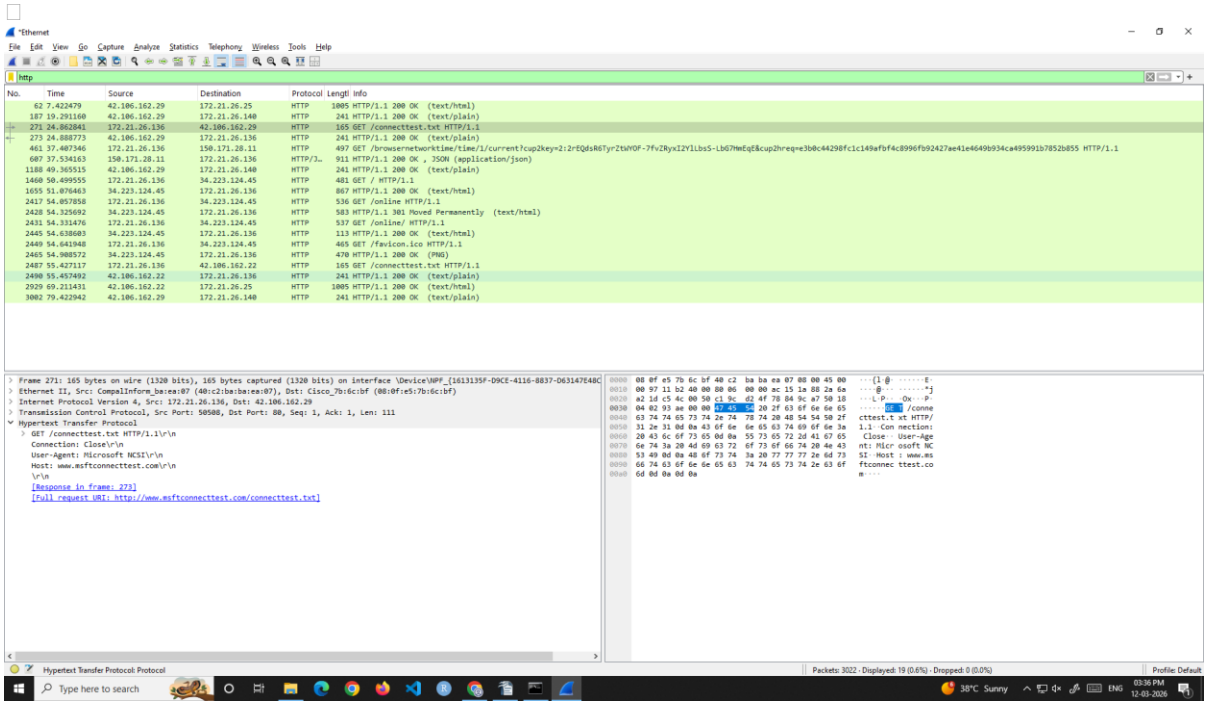
**Question 9:** Click on the HTTP GET request. In the Packet Details pane, expand the 'Hypertext Transfer Protocol' layer. Take a screenshot. What Host did your browser request?

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**Your Answer / Observation:**

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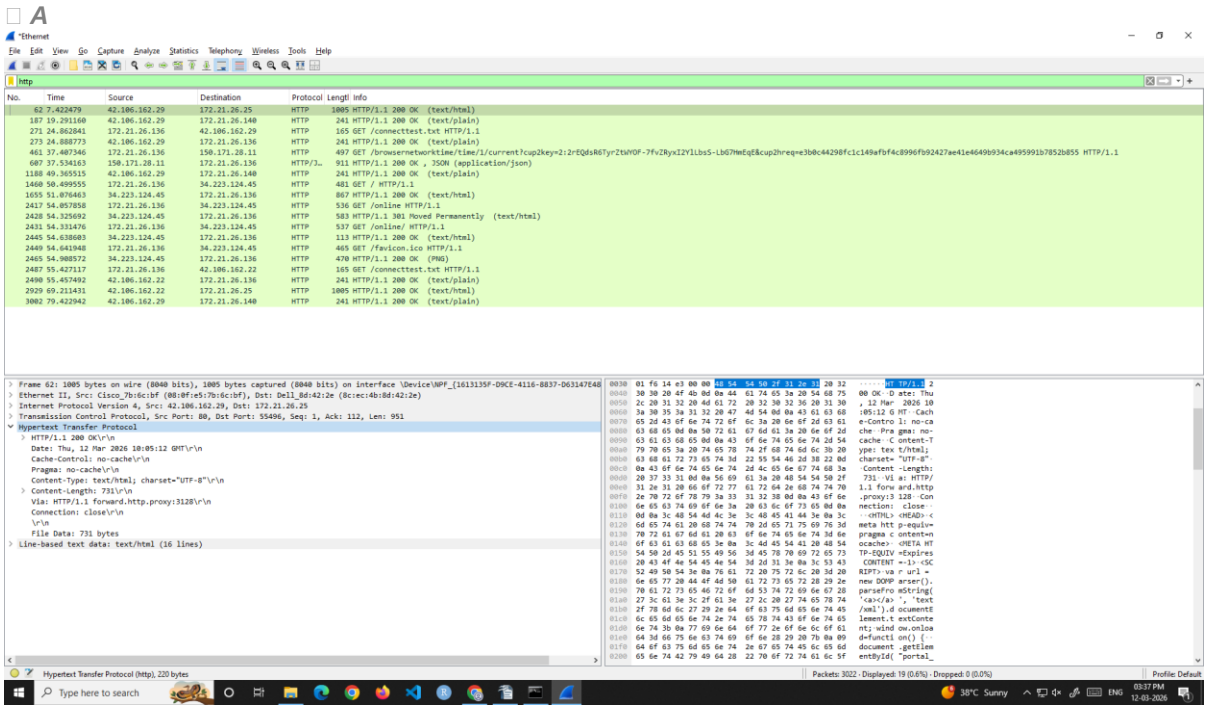
**Question 10:** Find an HTTP/1.1 200 OK response packet. Click on it and expand the HTTP layer. Take a screenshot. What is the Content-Type of the response?

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**Your Answer / Observation:**

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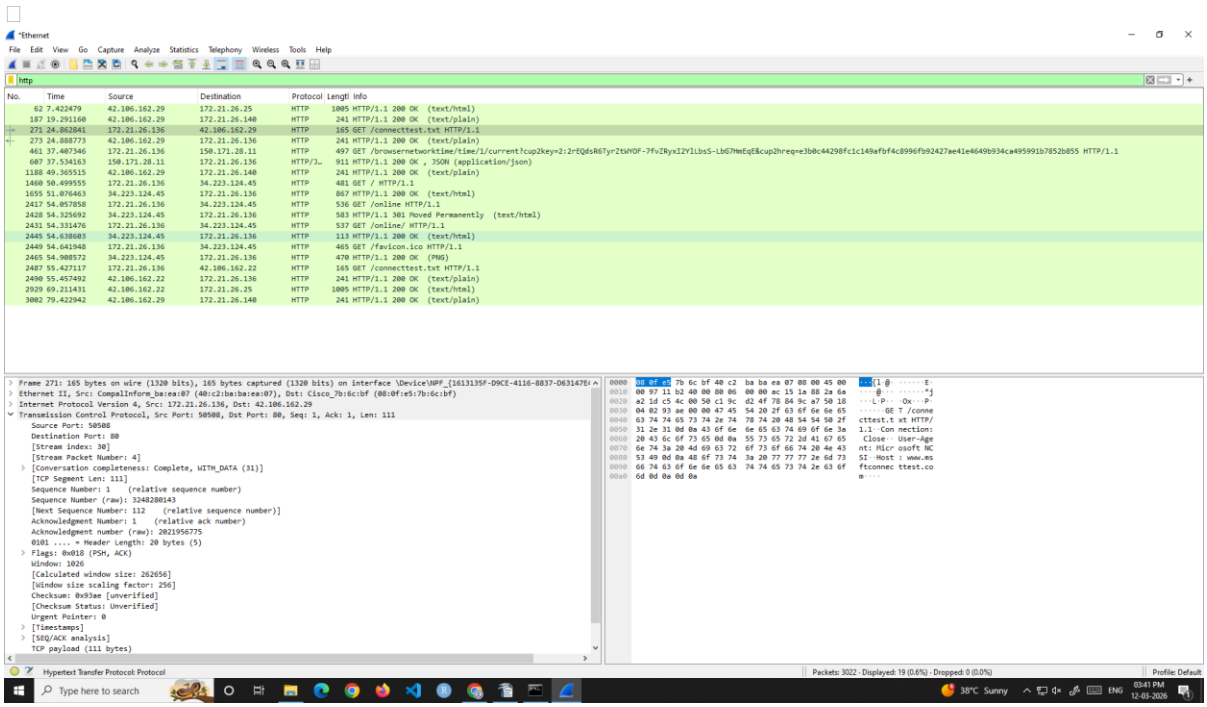
**Question 11:** In the Packet Details pane for the GET request, expand the 'Transmission Control Protocol' layer. Take a screenshot. What is the destination port number? What does this port number

tell you about the protocol?

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**Your Answer / Observation:**

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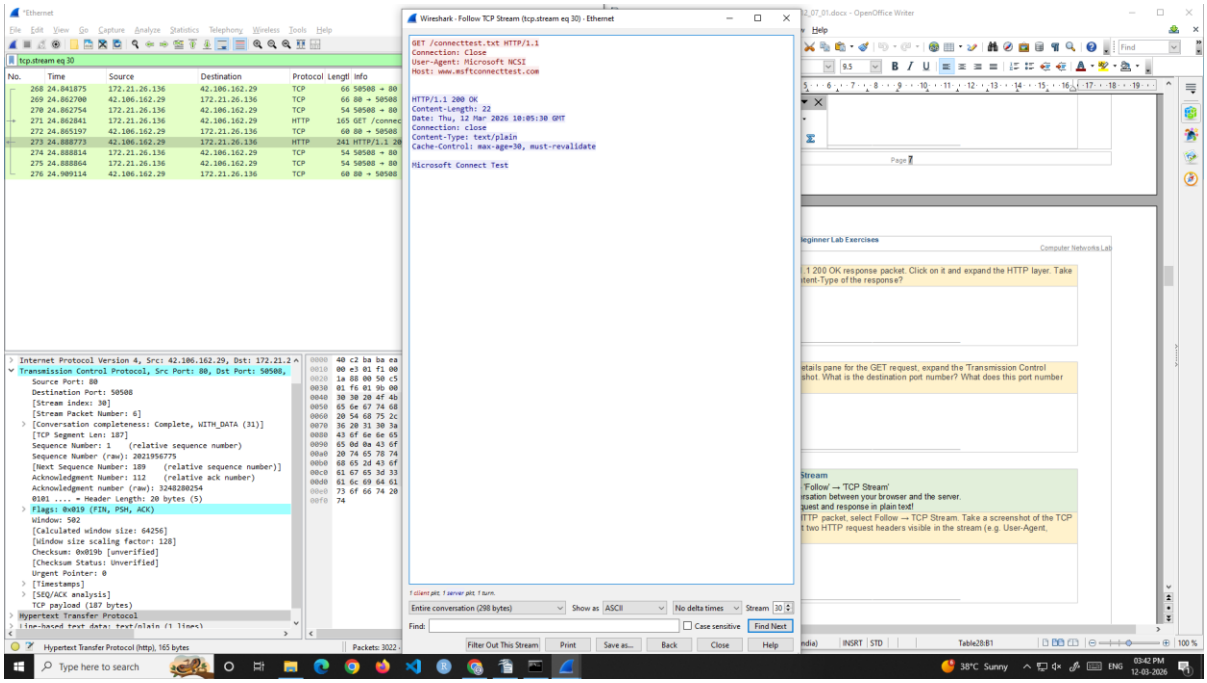
### How to Follow an HTTP Stream

Right-click any HTTP packet → 'Follow' → 'TCP Stream'
This reassembles the full conversation between your browser and the server.
You can read the raw HTTP request and response in plain text!

**Question 12:** Right-click an HTTP packet, select Follow → TCP Stream. Take a screenshot of the TCP Stream dialog. Identify at least two HTTP request headers visible in the stream (e.g. User-Agent, Accept).

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**Your Answer / Observation:**

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# Exercise 3 — DNS (Domain Name System)

### What is DNS?

DNS (Domain Name System) translates human-readable domain names (like www.google.com) into IP addresses (like 142.250.80.36) that computers can use.
Every time you visit a website, your computer first sends a DNS query to a DNS server to look up the domain name.
DNS typically uses UDP port 53 (or TCP port 53 for large responses).

DNS record types:

- A Record — maps a domain to an IPv4 address
- AAAA Record — maps a domain to an IPv6 address
- CNAME — alias from one domain to another
- MX — mail server for a domain

## Steps — Capturing DNS Traffic

**1** **Start a new capture in Wireshark**
Start capturing on your active network interface.

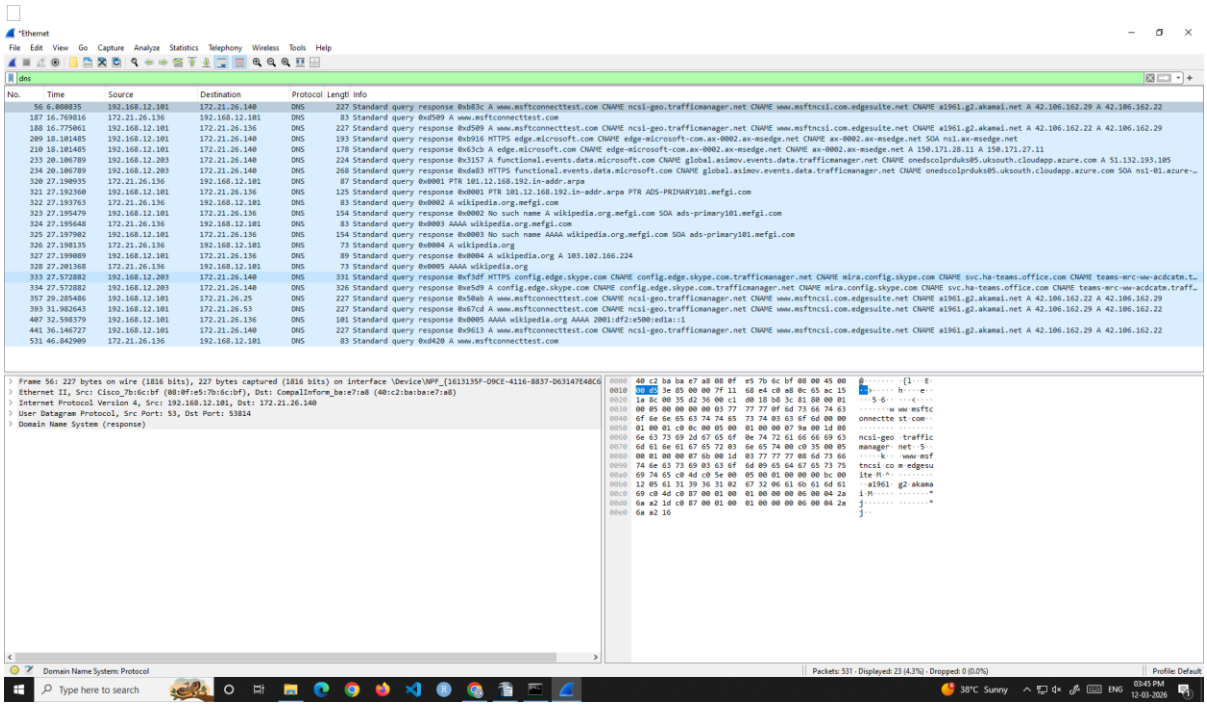
**2** **Open a terminal and run the nslookup command**
Type: nslookup wikipedia.org  then press Enter.Or open a browser and visit a brand new website you haven't visited today.

**3** **Stop the capture**
Return to Wireshark and stop capturing.

**4** **Apply the dns filter**
Type dns in the filter bar and press Enter.

## Questions — DNS Analysis

**Question 13:** Apply the dns filter and take a screenshot of the Packet List pane. You should see DNS Query and DNS Response packets. How many packets are shown?

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**Your Answer / Observation:**

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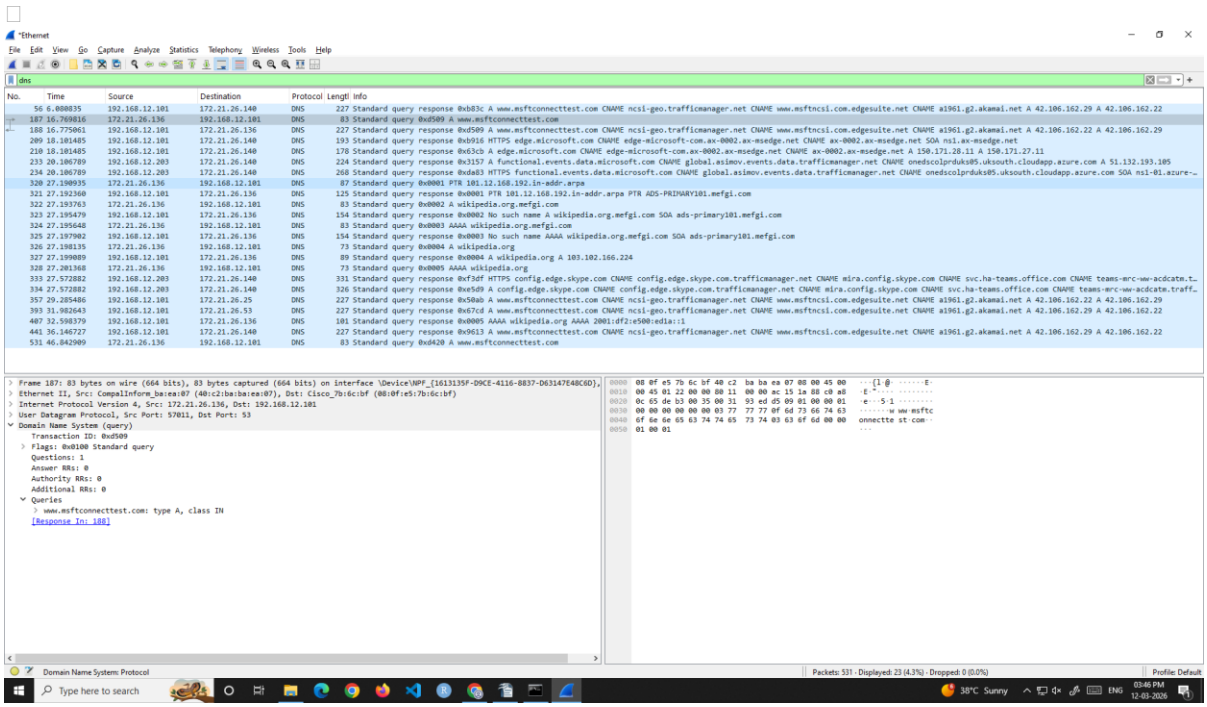
**Question 14:** Click on a DNS query packet. In the Packet Details pane, expand 'Domain Name

System (query)'. Take a screenshot showing the 'Queries' section. What domain name was queried?

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**Your Answer / Observation:**

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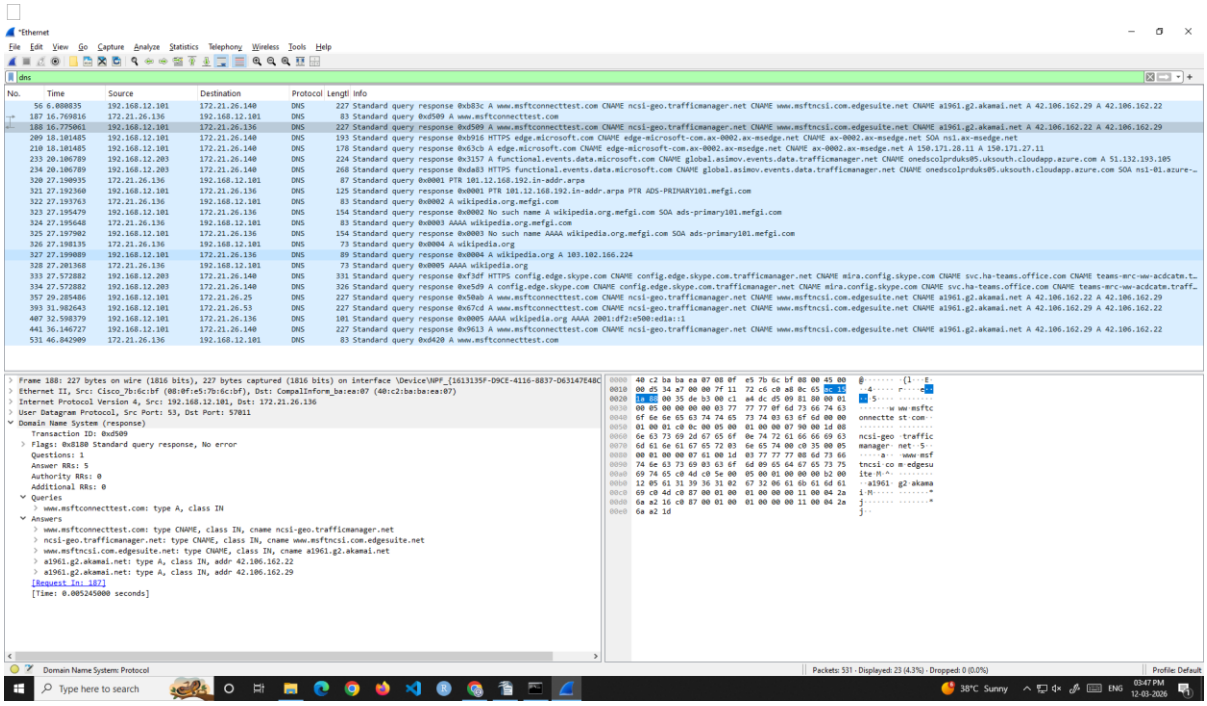
**Question 15:** Click on the corresponding DNS response packet. Expand 'Domain Name System (response)' → 'Answers'. Take a screenshot. What IP address was returned for the domain?

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**Your Answer / Observation:**

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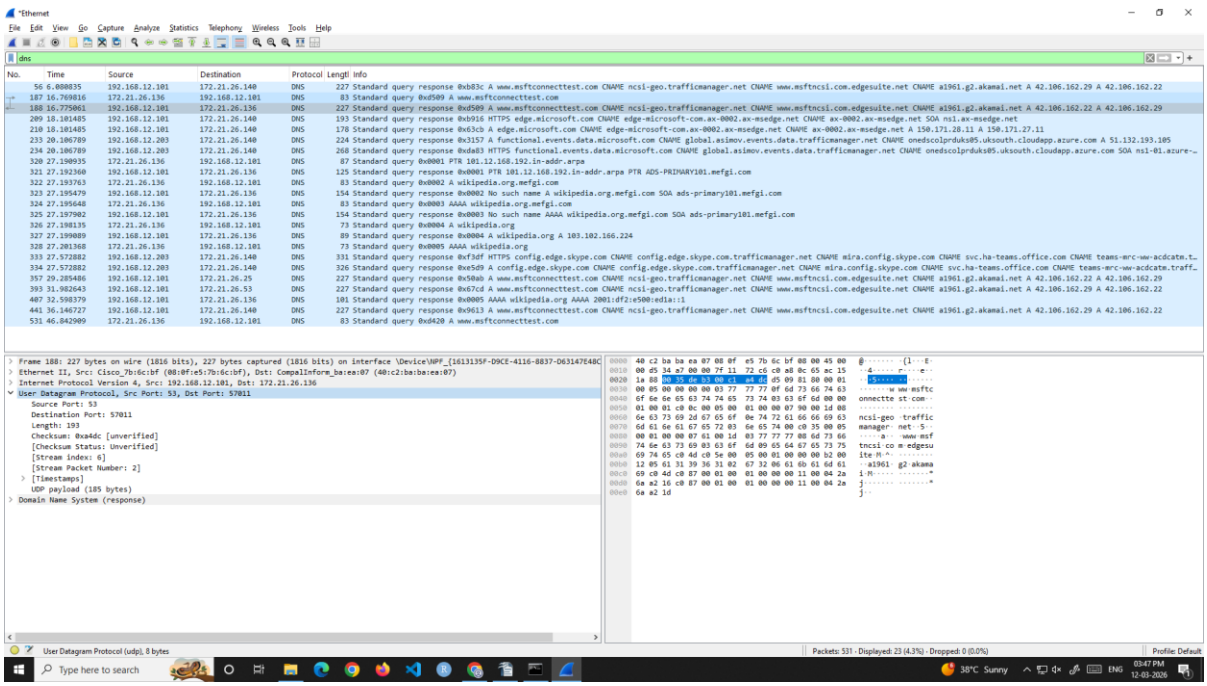
**Question 16:** What protocol and port number does DNS use? Look at the 'User Datagram Protocol' layer in the Packet Details pane and take a screenshot showing the source and destination port numbers.



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**Your Answer / Observation:**

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### Useful DNS Filter Tips

dns.qry.name == "google.com" — filter DNS queries for a specific domain
dns.flags.response == 0 — show only DNS queries
dns.flags.response == 1 — show only DNS responses

# Exercise 4 — TCP (Transmission Control Protocol)

### What is TCP?

TCP (Transmission Control Protocol) provides reliable, ordered, and error-checked delivery of data between applications.
Before data is exchanged, TCP establishes a connection using a Three-Way Handshake:

Step 1: Client sends SYN (Synchronize)
Step 2: Server sends SYN-ACK (Synchronize + Acknowledge)
Step 3: Client sends ACK (Acknowledge)

TCP operates at the Transport Layer (Layer 4) of the OSI model.
TCP flags: SYN | ACK | FIN | RST | PSH | URG

## Steps — Observing the TCP Three-Way Handshake

**1** **Start a new capture in Wireshark**
Start capturing on your active network interface.

**2** **Visit a website**
Open your browser and go to http://neverssl.com (use HTTP, not HTTPS, so the traffic is visible).

**3** **Stop the capture**
Return to Wireshark and stop capturing.

**4** **Filter for TCP traffic to port 80**
In the filter bar, type: tcp.port == 80 and press Enter.

**5** **Find the TCP handshake**
Look for the sequence: [SYN] → [SYN, ACK] → [ACK] at the start of the packet list.

## Questions — TCP Analysis

**Question 17:** Apply the filter tcp.port == 80 and take a screenshot. Identify the three packets of the TCP three-way handshake (SYN, SYN-ACK, ACK) in the Info column.

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 18:** Click on the [SYN] packet. Expand 'Transmission Control Protocol' in the Packet Details pane. Take a screenshot. What is the Sequence Number of this first SYN packet?

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 19:** Click on the [SYN, ACK] packet sent by the server. Take a screenshot showing the TCP flags section. Which flags are set to 1 (enabled)?

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 20:** Look at the Source Port in one of the TCP packets from your computer. Take a screenshot. Port numbers above 1024 are 'ephemeral ports' assigned by your OS. Write down the ephemeral port your computer used for this connection.

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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### ☐ TCP vs UDP — Key Difference

TCP: Connection-oriented, reliable, ordered. Uses handshake. Example: HTTP, FTP, SSH.
UDP: Connectionless, faster but no delivery guarantee. Example: DNS, video streaming, gaming.

# Exercise 5 — ARP (Address Resolution Protocol)

### What is ARP?

ARP (Address Resolution Protocol) maps an IP address to a physical MAC address on a local network. When your computer wants to communicate with another device on the same network, it sends a broadcast ARP Request asking 'Who has IP address X? Tell me your MAC address.'
The device with that IP responds with an ARP Reply containing its MAC address.

ARP operates at the Data Link Layer (Layer 2) / Network Layer (Layer 3) boundary.
ARP packets are only relevant on local networks — they are NOT routed across the internet.

## Steps — Capturing ARP Traffic

**1** **Start a new capture in Wireshark**
Start capturing on your active network interface.

**2** **Generate ARP traffic**
Open a terminal and type: Windows: arp -d (clears ARP cache, forces new lookups) Mac/Linux: sudo arp -a -dThen run: ping 192.168.1.1 (or your router's IP address).

**3** **Stop the capture and apply the arp filter**
Type arp in the filter bar and press Enter.

## Questions — ARP Analysis

**Question 21:** Apply the arp filter and take a screenshot of the Packet List pane showing ARP Request and ARP Reply packets.

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 22:** Click on an ARP Request packet. In the Packet Details pane, expand 'Address Resolution Protocol (request)'. Take a screenshot. What Target IP Address is being asked about? What is the Target MAC address shown (hint: it should be all zeros)?

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 23:** Click on the ARP Reply packet. Expand the ARP layer. Take a screenshot. What MAC address was returned in the reply? (This is your router's MAC address.)

☐ ***Attach Screenshot Here***

**Your Answer / Observation:**

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**Question 24:** Look at the Destination address of the ARP Request packet in the Packet List pane. Take a screenshot. What destination MAC address is used for the broadcast? What does ff:ff:ff:ff:ff:ff mean?

☐ ***Attach Screenshot Here***

**Your Answer / Observation:**

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# Exercise 6 — Display Filters Challenge

### Wireshark Display Filter Cheat Sheet

Protocol filters: http | icmp | dns | tcp | udp | arp

IP address filters:
- ip.addr == 8.8.8.8 — packets to or from this IP
- ip.src == 192.168.1.5 — packets FROM this IP
- ip.dst == 192.168.1.1 — packets TO this IP

Port filters:
- tcp.port == 80 — TCP traffic on port 80
- udp.port == 53 — UDP traffic on port 53

Combining filters:
- http && ip.src == 192.168.1.5 — HTTP packets from a specific IP
- icmp || arp — ICMP OR ARP packets
- !http — everything EXCEPT HTTP

## Challenge Questions

For each question below, write the display filter you used, then attach a screenshot.

**Question 25:** Write a display filter to show ONLY DNS packets that are responses (not queries). Hint: use dns.flags.response. Write the filter in the answer box and attach a screenshot.

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 26:** Write a display filter to show HTTP traffic from your own IP address only (use ip.src with your IP). Attach a screenshot showing the result.

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 27:** Capture some general traffic. Apply a filter to show ALL traffic EXCEPT ARP and DNS. Write the filter you used and attach a screenshot.

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 28:** Use the filter tcp.flags.syn == 1 && tcp.flags.ack == 0 to find ONLY TCP SYN packets (connection initiation). Take a screenshot. How many SYN packets are in your capture?

☐ ***Attach Screenshot Here***

**Your Answer / Observation:**

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# Exercise 7 — Statistics and Protocol Hierarchy

### Wireshark Statistics Tools

Wireshark includes built-in tools to summarize capture data:

- Statistics → Protocol Hierarchy — shows which protocols are in your capture and what % of traffic they represent
- Statistics → Conversations — shows all unique conversations (by IP, TCP, UDP)
- Statistics → IO Graph — shows traffic volume over time as a graph
- Statistics → Endpoints — lists all unique IP and MAC addresses seen

## Steps

**1** **Capture general traffic for 30 seconds**
Browse 2-3 websites, run a ping, and let DNS resolve some domains. Then stop the capture.

**2** **Open Protocol Hierarchy**
Go to Statistics menu → Protocol Hierarchy.

**3** **Open IO Graph**
Go to Statistics menu → IO Graph.

## Questions — Statistics

**Question 29:** Open Statistics → Protocol Hierarchy. Take a screenshot of the Protocol Hierarchy window. Which protocol had the highest percentage of packets in your capture?

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 30:** Open Statistics → IO Graph. Take a screenshot of the IO Graph. Describe in 1-2 sentences what the graph shows about the traffic pattern during your capture.

☐ *Attach Screenshot Here*

**Your Answer / Observation:**

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**Question 31:** Open Statistics → Conversations → TCP tab. Take a screenshot. Which IP address appears most frequently as a remote endpoint? What service do you think this represents?

☐ ***Attach Screenshot Here***

**Your Answer / Observation:**

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# Part 8 — Summary and Reflection

| Protocol | Layer (OSI) | Transport | Port | Purpose |
|---|---|---|---|---|
| **HTTP** | Layer 7 – Application | TCP | 80 | Web page transfer |
| **HTTPS** | Layer 7 – Application | TCP | 443 | Encrypted web transfer |
| **DNS** | Layer 7 – Application | UDP / TCP | 53 | Domain name resolution |
| **ICMP** | Layer 3 – Network | N/A | N/A | Ping / error messages |
| **TCP** | Layer 4 – Transport | — | Varies | Reliable data delivery |
| **UDP** | Layer 4 – Transport | — | Varies | Fast, unreliable delivery |
| **ARP** | Layer 2 / 3 | N/A | N/A | IP to MAC resolution |

## Reflection Questions

Answer the following questions in your own words (no screenshot required).

**Question 32:** In your own words, explain what happens step-by-step when you type a URL like www.google.com in your browser and press Enter. Include DNS, TCP, and HTTP in your explanation.

**Question 33:** Why can Wireshark see ICMP and HTTP traffic but NOT HTTPS traffic in readable form? What technology prevents you from reading HTTPS packets?

**Question 34:** What is the difference between a Display Filter and a Capture Filter in Wireshark? When would you use each?

**Question 35:** Give one real-world use case where a network administrator might use Wireshark. Explain what they are looking for and why packet analysis is useful.

## Lab Complete!

Instructor / Teaching Assistant Signature: _____

Grade: _____ / 35 Questions Date Checked: _____